KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 1

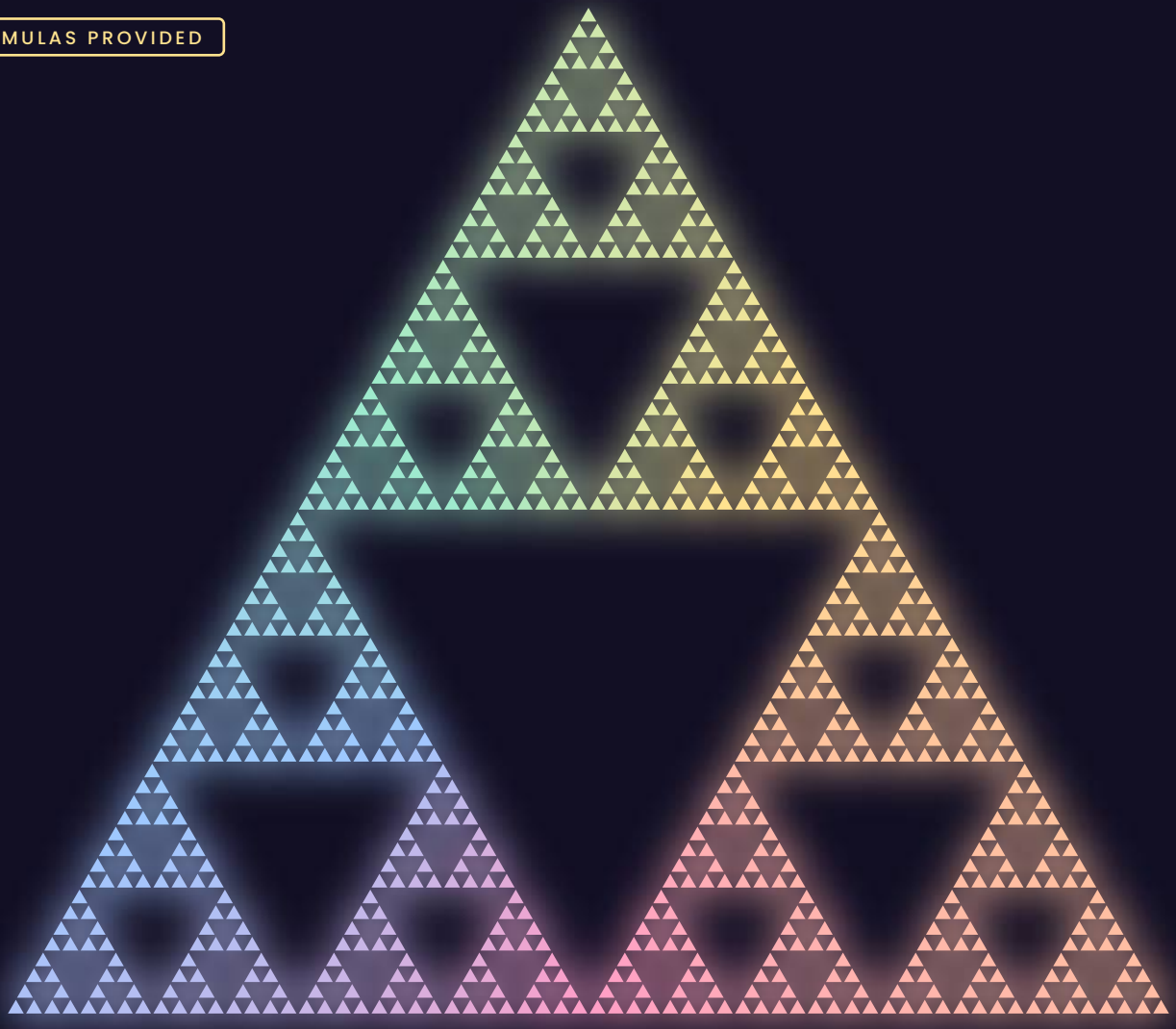
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 10

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

SIERPIŃSKI TRIANGLE
three copies of itself, each half the size

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

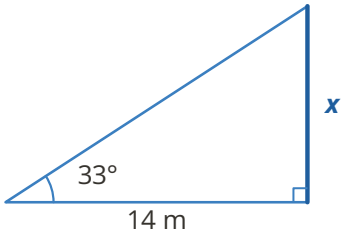
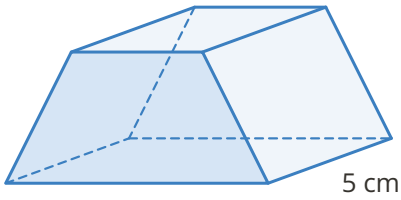
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 24 cm^2. Find the volume of the prism. 	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find Leo's commission. Sales: $2500 Commission rate: 10%	FORMULA Commission = percentage × value of sales COPY SUBSTITUTE ANSWER
4 Share $56 in the ratio 3 : 4.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is white. Colour: Red, White Marbles: 5, 3	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

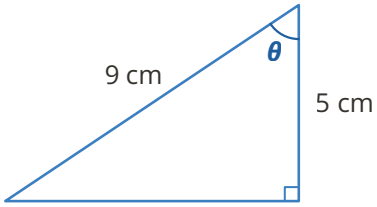
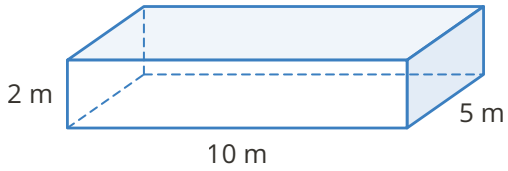
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA $\theta = \sin^{-1}\frac{O}{H}$, $\cos^{-1}\frac{A}{H}$ or $\tan^{-1}\frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price: \$300 Selling price: \$345	FORMULA Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 Simplify. $18k^7 \div 6k^3$	FORMULA $a^m \div a^n = a^{m-n}$ COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 5$. $y = 2x - 3$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

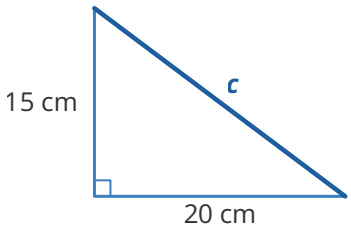
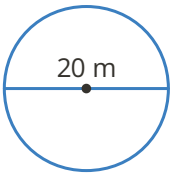
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the new population of the town. Population 2400 Increase 5%	FORMULA New amount = original × (100% + increase) COPY SUBSTITUTE ANSWER
4 Find x when $y = 23$, $m = 4$ and $c = -1$. $y = mx + c$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER
5 Find the probability that a goal is not scored. $P(\text{goal}) = \frac{2}{5}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 2

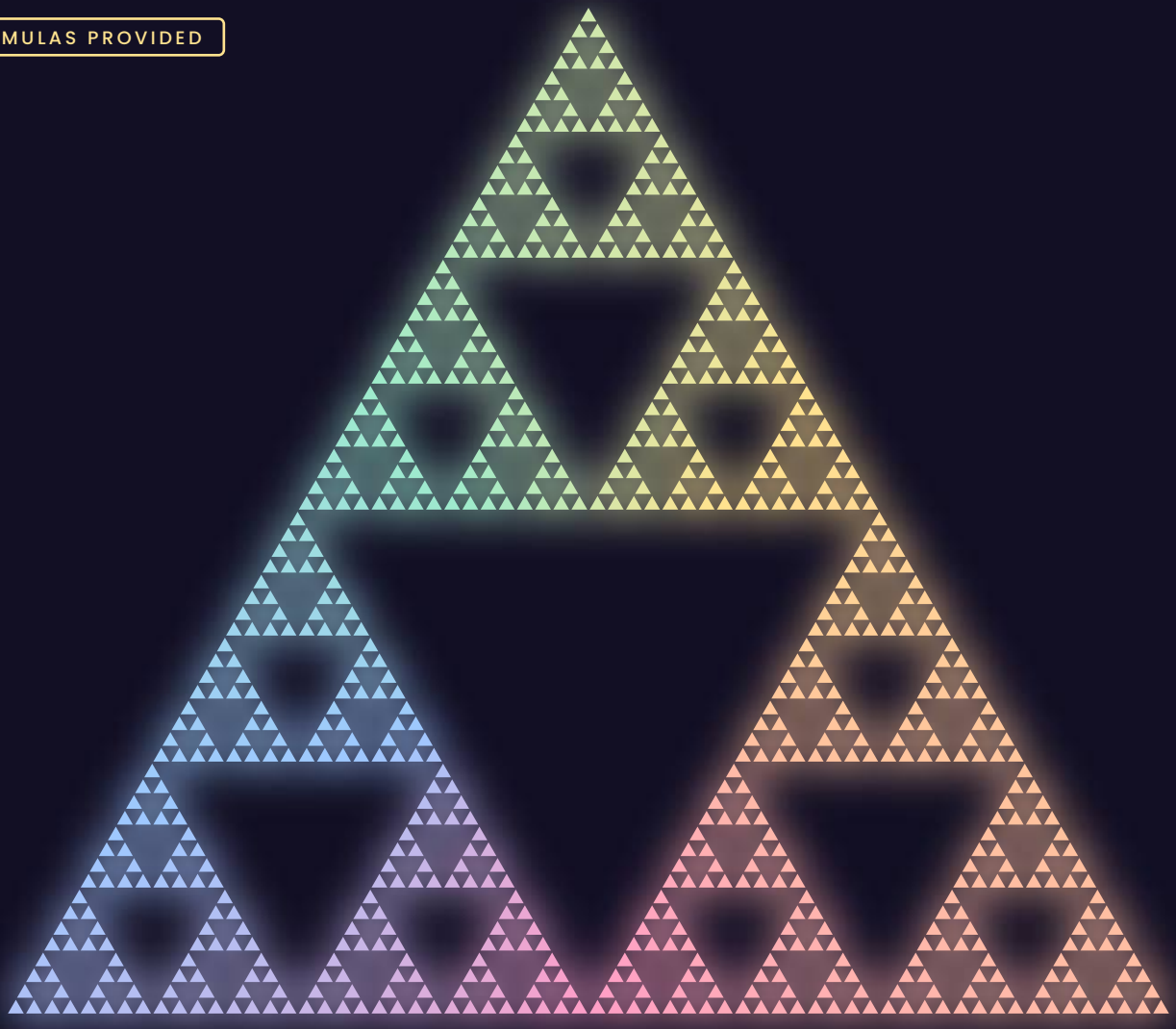
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ACHIEVEMENT THROUGH ENDEAVOUR

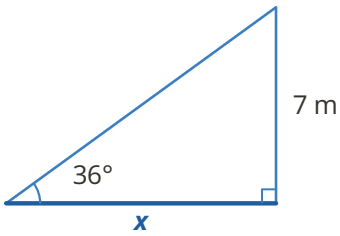
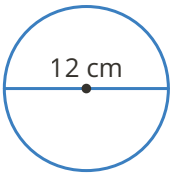
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the new price. Original price: $150 Increase: 30%	FORMULA New amount = original × (100% + increase) COPY SUBSTITUTE ANSWER
4 Simplify. $(m^3)^4$	FORMULA $(a^m)^n = a^{mn}$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is white. Colour: Red, White Marbles: 5, 3	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

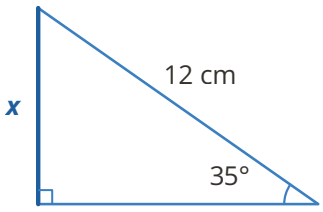
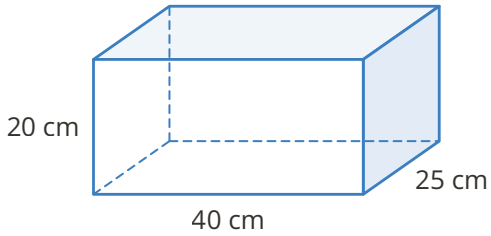
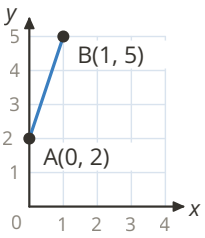
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 How many litres does this tank hold?  	FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$ COPY SUBSTITUTE ANSWER
3 Find the simple interest on \$2000 invested at 4% per year for 3 years.	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 Evaluate. Write your answer as a fraction. 10^{-1}	FORMULA $a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

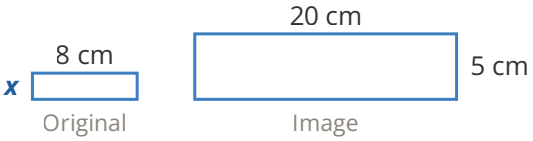
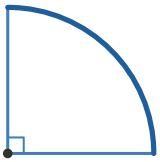
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find the value of x. 	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place. 	FORMULA $l = \frac{\theta}{360} \times 2\pi r$ COPY SUBSTITUTE ANSWER
3 Find the original price of the pair of shoes. Discount: 25% Sale price: $90	FORMULA Original amount = known amount ÷ its percentage × 100 COPY SUBSTITUTE ANSWER
4 Write as a basic numeral. 3.2×10^4	FORMULA $a \times 10^n$, where $1 \leq a < 10$ COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 2$. $y = 4x - 1$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 3

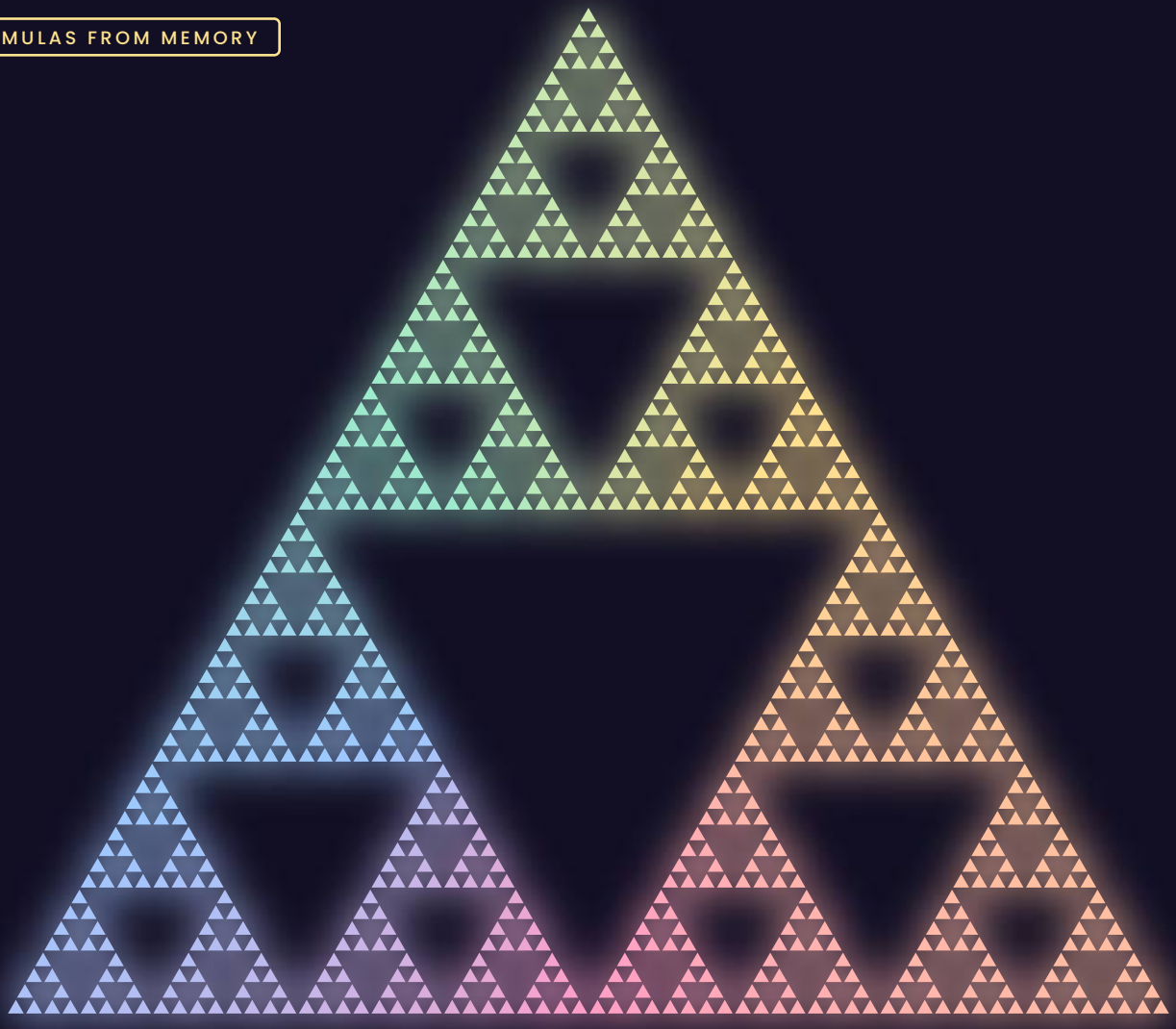
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FORMULAS FROM MEMORY



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ACHIEVEMENT THROUGH ENDEAVOUR

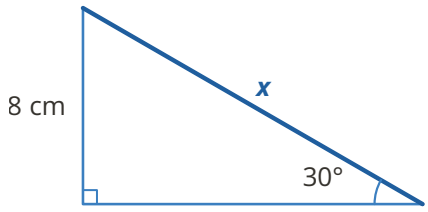
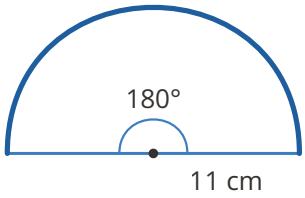
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Josh earns a salary of $57 200 a year. Find Josh's weekly pay.	FORMULA SUBSTITUTE ANSWER
4 Write as a single power. $2^3 \times 2^4$	FORMULA SUBSTITUTE ANSWER
5 State the gradient and the y-intercept. $y = -x + 3$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

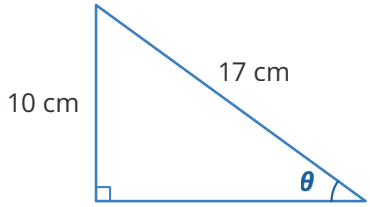
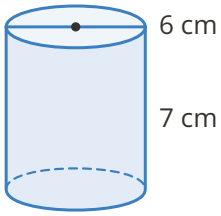
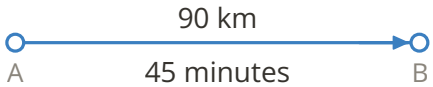
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
3 Find the percentage profit. Cost price: $300 Selling price: $345	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
4 A car travels from A to B. Find the average speed in km/h.  	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
5 Find the mean of these scores. Scores: 17, 23, 20, 26, 19	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

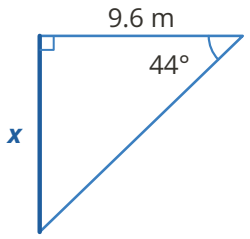
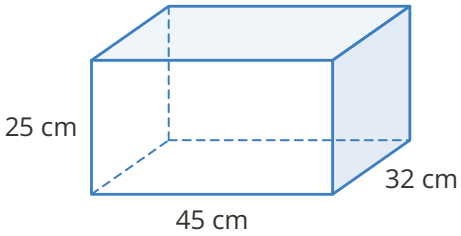
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 How many litres does this tank hold?  	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest on \$2000 invested at 4% per year for 3 years.	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 72 500 000	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the player does not win. $P(\text{win}) = 0.47$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 4

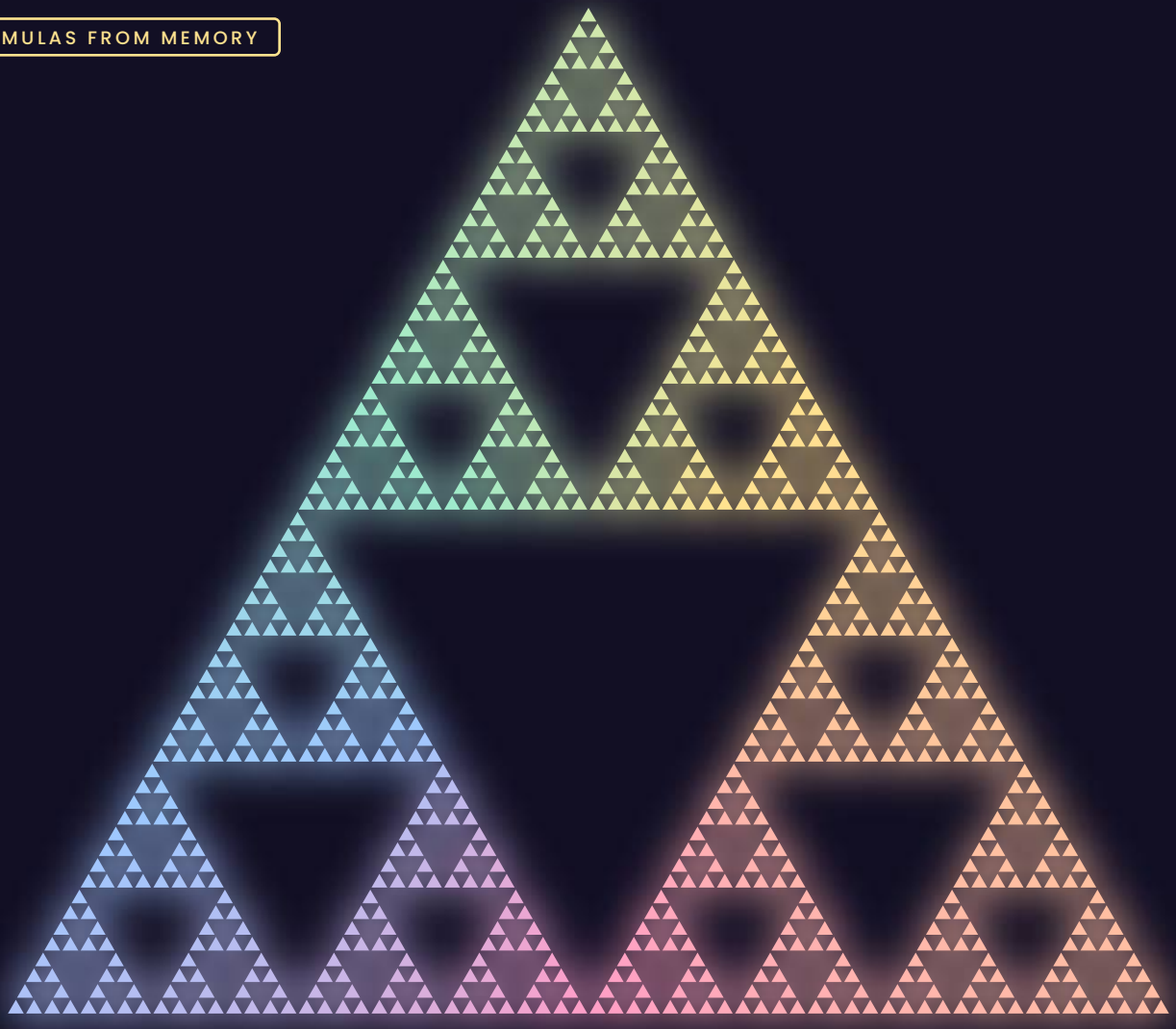
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ACHIEVEMENT THROUGH ENDEAVOUR

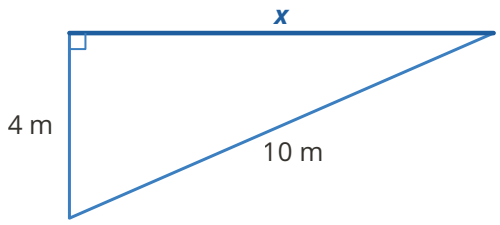
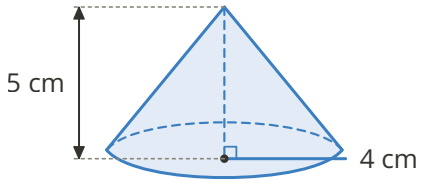
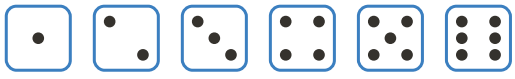
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cone, correct to 1 decimal place. 	FORMULA $V = \frac{1}{3}\pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find Zoe's total pay for the week. Hourly rate \$28 Normal hours 38 Overtime hours (time-and-a-half) 6	FORMULA SUBSTITUTE ANSWER
4 Find x when $y = -7$, $m = 2$ and $c = 3$. $y = mx + c$	FORMULA SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a number less than 3. 	FORMULA SUBSTITUTE ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

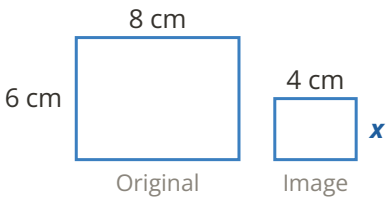
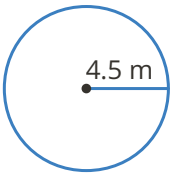
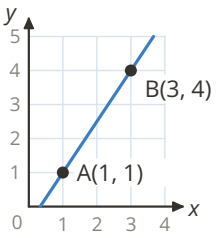
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find the value of x.  	FORMULA SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price: $36 Selling price: $45	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 607 000	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

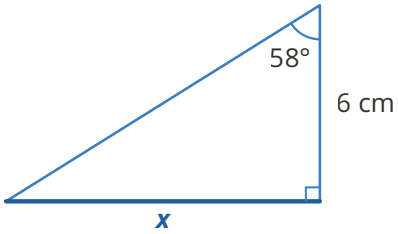
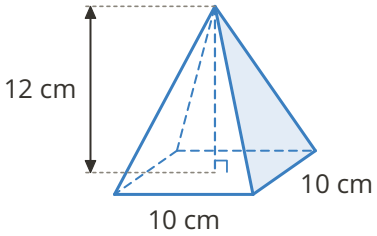
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this pyramid. 	FORMULA $V = \frac{1}{3}Ah$ (A = area of the base) COPY SUBSTITUTE ANSWER
3 Chloe earns a salary of \$66 650 a year. Find Chloe's fortnightly pay, to the nearest cent.	FORMULA SUBSTITUTE ANSWER
4 Simplify. $(5x)^0$	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these lengths. Metres: 2.4, 3.1, 2.8, 3.5, 3.2	FORMULA SUBSTITUTE ANSWER